

Data cleaning passport

Before VS After

1. Dataset Comparison: Before and After cleaning

Characteristic	Before cleaning (anime_catalog.csv)	After cleaning (anime_catalog_clean.csv)
Number of rows	10 142	10 142
Number of columns	26	16
Missing values	Yes (there are more than 20% omissions in 10 columns)	No (all omissions have been processed)
Year format	String format (with text, e.g. "Весна 1983")	Integer (e.g. 1983)
The format of the views (site_views)	String (e.g. "27,311 views")	Integer (e.g. 27311)
Lists (genres, studios, etc.)	Merged string with a separator " "	Structured lists (Python lists)

2. Preprocessing Steps

1. Analyzing and deleting uninformative columns:

During the initial inspection, 10 columns were identified, in which more than 20% of the data was missing. Of these, 8 columns had 100% omissions (for example, shikimori_rating, mal_rating) and were completely removed from the dataset. The kinopoisk_rating and kinopoisk_url columns were also deleted due to the large number of omissions and low relevance for further analysis.

2. **Handling missing values (Imputation):**

- In the alt_names column (alternative names), single omissions have been replaced with empty rows.
- Omissions in the categorical text attributes genres, director, dubbing, and studio were filled in with the 'Unknown' stub to preserve the data structure.
- In the year column, the method of extracting a 4-digit number from the text was applied. The remaining gaps (where the year was not initially specified) were filled in with the median value of the year for the entire dataset, after which the data type was converted to an integer (int).

3. **Data type conversion:**

Only numeric values were extracted from the site_views column (number of views) using regular expressions, and textual garbage was removed. The cleared values are converted to the int numeric format for the possibility of quantitative analysis.

4. **Formatting string arrays:**

The values in the genres, alt_names, studio, director, and dubbing columns, originally written as a single text with a | separator, were programmatically separated. These cells now contain complete lists, which greatly simplifies data filtering and aggregation.